

MAZAK VQC-20/40

LinuxCNC / Mesa Retrofit — Complete Manual Set

Machine: Mazak Vertical Quality Center VQC-20/40 | **S/N:** 060231
Original control: Mazatrol M-2 (retired) | **New control:** LinuxCNC + Mesa 7i97T / 7i49 / 7i84U
Shop power: 460 VAC 3-phase (onboard transformer tapped for 460 V)
Document: MZK-LCNC-000 — Master Guide | **Revision:** A — July 2026

Volume 0 — Master Guide & Table of Contents

How This Manual Set Is Organized

This set documents the Mazak VQC-20/40 (SN 060231) **as converted** from its original Mazatrol M-2 control to LinuxCNC with Mesa Electronics FPGA hardware. It supplements — it does not replace — the original Mazak factory manuals, which remain authoritative for the base machine's mechanics, lubrication, and original electrical drawings.

Volume	Document No.	Title	Audience
0	MZK-LCNC-000	Master Guide & Table of Contents	Everyone
1	MZK-LCNC-001	Machine Overview & Safety	Everyone — read first
2	MZK-LCNC-002	Operation Manual (LinuxCNC)	Operators
3	MZK-LCNC-003	Mesa Conversion & Installation	Integrator / electrician
4	MZK-LCNC-004	Wiring Diagrams & I/O Reference	Integrator / electrician
5	MZK-LCNC-005	Maintenance & Troubleshooting	Maintenance
6	MZK-LCNC-006	Reference, Glossary & Index	Everyone

⚠ PROJECT STATUS NOTICE. As of this revision the retrofit is in the **planning / bring-up stage**. Most wiring assignments in Volume 4 carry a **PLAN_VERIFY** status: they are engineered plans that **must be traced and confirmed in the cabinet** before wires are landed or outputs energized. Statuses are printed with every pin table. Nothing in this set overrides the verification steps in Volume 3.

Master Table of Contents

Volume 1 — Machine Overview & Safety

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- 2. Key Project Files (repo & OneDrive paths)
- 3. External References
- 4. Glossary of Terms
- 5. Alphabetical Index (full set)

Source Documents

This manual set was compiled July 2026 from these authoritative project sources:

Source	Location
Original Mazak manuals (11 PDFs)	OneDrive–Personal/OneDrive Docs/Mazak/Manuals_SN060231/
Architecture decision	GitHub/Github LinuxCNC/docs/architecture_decision.md
Project status & TODO	GitHub/Github LinuxCNC/docs/project_status.md
Current pin authority (91 signals)	GitHub/Github LinuxCNC/mesa/current_pin_authority.csv
Wiring master workbook	GitHub/Github LinuxCNC/wiring/Mazak_Wiring_Master_7i97T_7i49_7i84U.xlsx
HAL / INI skeletons	GitHub/Github LinuxCNC/linuxcnc/
PLC signal dictionary (385 signals)	Manuals_SN060231/02_Maintenance/YM2V39L_element_list.csv

End of Volume 0.

Volume 1 — Machine Overview & Safety

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

Document MZK-LCNC-001 · Rev A · July 2026

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Chapter 1 — About the Machine

The Mazak **Vertical Quality Center VQC-20/40** is a vertical machining center originally delivered with a **Mazatrol M-2** conversational CNC control (the 41434WB schematic set is titled M-1 because it is the original 10/1983 drawing set; the NC on SN 060231 was upgraded to M-2).

Item	Detail
Model	Mazak VQC-20/40
Serial number	060231
Original control	Mazatrol M-2 (Meldas-based)
Axes	X, Y, Z — analog velocity-command servo drives (Mitsubishi TRA type)
Servo motors	Mitsubishi HD81-12S class, tachometer velocity feedback
Position feedback	Tamagawa resolvers (documented: RT-5XA-11, RT-5XA-L1; TS2014N noted per axis in pin authority)
Spindle drive	Mitsubishi FR-SX with analog speed reference
Spindle gearbox	Hi/Lo hydraulic shift (SOL-12 / SOL-13, PRS-10 / PRS-12 confirm)
ATC	Hydraulic tool changer + BCD-coded magazine (PRS-21...25)
Tooling	Per Mazak Tooling Manual M00T002 (BT/CAT 40-50 family)
Incoming power	460 VAC 3-phase; onboard variable-voltage transformer tapped for 460 V, stepping to the 230 VAC internal bus. No external transformer or phase converter is required.
Control power	24 VDC bus from HR-11F-24 supply (P24/G24 distribution)

Chapter 2 — The Conversion at a Glance

The retrofit replaces **only the Mazatrol M-2 NC unit** (the *NUMERICAL CONTROL UNIT* block on schematic sheet 4143075312). Everything below stays:

Retained: X/Y/Z servo amplifiers, servo motors and tachometers, Tamagawa resolvers, FR-SX spindle drive, hydraulic power unit and ATC, magazine, solenoids and prox switches, 24 VDC control bus, cabinet terminal strips, transformer and power distribution.

Replaced / Added:

Old (Mazatrol M-2)	New (LinuxCNC/Mesa)
Mazatrol NC unit & CRT/keyboard	LinuxCNC control PC + monitor
NC-to-machine interface	Mesa 7i97T Ethernet analog servo controller (static IP 192.168.1.121, hm2_eth)
Resolver-to-NC feedback path	Mesa 7i49 resolver-to-digital interface (plain, 5 kHz excitation baseline)
Mazatrol PLC I/O rack	Mesa 7i84U remote field I/O on smart-serial, near the green breakout PCB
Mazatrol conversational programming	Standard G-code (plus optional WHB04B USB pendant)

The original PLC ladder logic (manual YM2V39L, 385 catalogued signals) defines the interlock behavior that the LinuxCNC/HAL configuration must reproduce.

Chapter 3 — General Safety Rules

1. **Only trained personnel** may operate, wire, or service this machine.
2. **Read Volume 3 before energizing anything.** During bring-up, most I/O assignments are unverified plans.
3. Never bypass, jumper, or software-defeat a safety interlock — door switch, overtravel limit, or E-stop.
4. Keep guards and doors closed during spindle rotation and program execution.
5. Know where every E-stop is before you need one.
6. No loose clothing, jewelry, or unsecured long hair near the spindle or ATC.
7. Verify workholding and tool retention (correct pull studs per M00T002) before cycle start.
8. The ATC and magazine move with hydraulic force — stay clear of the tool-change envelope whenever hydraulics are pressurized.

Chapter 4 — The Safety Chain (E-stop Philosophy)

Rule 1 of this retrofit: a **hardware** safety chain must remove hazardous power. LinuxCNC and HAL logic **supplement** the chain — they are never the only thing standing between an E-stop button and moving iron.

- The E-stop string (buttons, door interlocks, overtravel finals, servo contactor) is wired in hardware. Pressing E-stop drops the servo/hydraulic contactors **directly**, regardless of the PC.

- The Mesa 7i97T monitors the chain on TB5.10 (estop-ext) so LinuxCNC *knows* the chain opened and can abort motion cleanly — monitoring, not actuation.
- Original chain wiring must be **traced and recorded before any control rewiring** (see Volume 3, Ch. 9).
- After any change to the chain, perform the full E-stop functional test in Volume 3 §9 before production use.

Chapter 5 — Lockout / Tagout & Electrical Safety

- **460 VAC 3-phase** enters the cabinet; the transformer, drive bus, and spindle drive remain lethal after the door opens. Lock out at the wall disconnect and verify dead before work.
- Drive DC bus capacitors hold charge after power-off — wait and measure before touching drive terminals.
- The 24 VDC field bus (HR-11F-24) powers solenoids and relays. A "safe" 24 V circuit can still fire a hydraulic valve — treat it with respect during ATC work.
- One person, one lock. Test-before-touch with a meter you have proven on a known-live source.

Chapter 6 — Retrofit-Specific Hazards

Hazard	Why it exists	Defense
Axis runaway	A reversed analog command polarity makes the servo run away at full speed the moment it is enabled.	Bring-up procedure enables one axis at a time, drives inhibited first, zero-command verified, low gain (Vol 3 §8).
Resolver mis-wiring	Original Meldas/TRA wiring may drive the resolver "backwards" (excitation into stator). The 7i49 expects the opposite.	Ohmmeter the winding pairs before power ; never trust original wire names (Vol 4 §3).
Dual excitation	If any legacy circuit still drives the resolver windings while the 7i49 is connected, signals and hardware can be damaged.	The 7i49 must be the sole excitation source — verified before energizing.
Unverified I/O states	NO/NC and active-high/low assumptions in the HAL files are placeholders. A wrong assumption can fire a solenoid on boot.	Every output stays disconnected until its normal state is measured (PLAN_VERIFY discipline, Vol 4 §10).
Over-current outputs	Solenoid and contactor coils (some 100 VAC) exceed Mesa output ratings.	Interposing relays RLY-1...7 with flyback/RC suppression (Vol 4 §§4, 9).
ATC motion faults	ATC forward/reverse outputs energized without proven interlocks can crash the changer.	ATC outputs are last in bring-up order, dry-run only, no tool load (Vol 3 §8, step 8).

End of Volume 1. Continue to Volume 2 — Operation Manual.

Volume 2 — Operation Manual (LinuxCNC)

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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Applicability: This volume describes operation **after** bring-up is complete and all Volume 3 verification checklists are signed off. During bring-up, follow Volume 3 procedures instead.

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Chapter 1 — Control Station Overview

The Mazatrol CRT and keyboard are replaced by a **LinuxCNC PC**. The machine configuration is `mazak_vqc_20_40.ini`, which loads the HAL files that map the Mesa hardware:

- `motion_7i97t.hal` — axes, spindle command, limits/homes/E-stop
- `field_7i84u.hal` — ATC, magazine, coolant, lube, hydraulics, panel I/O
- `postgui.hal` / `pendant_whb04b.hal` — GUI extras and optional pendant

Physical operator devices that may be retained on the original panel (wired to the 7i84U): **Cycle Start** (IN28), **Feed Hold** (IN29), **Single Block** (IN30), plus the alarm light/horn (OUT15). All are marked optional-reuse in the pin authority table.

Chapter 2 — Power-Up Sequence

1. Close the cabinet; verify no lockout tags.
2. Turn on the main disconnect (460 VAC). The transformer energizes the 230 VAC bus and the HR-11F-24 24 VDC supply.
3. Wait for the hydraulic pump if it is on a delay; confirm hydraulic pressure OK (`hydraulic-ok` , Sanwa SPS-8T-PC-20).
4. Power on the LinuxCNC PC. The 7i97T (192.168.1.121) is discovered over Ethernet at LinuxCNC start.
5. Launch LinuxCNC and select the `mazak_vqc_20_40` configuration.
6. Release/reset all E-stops, then press **F1 (E-stop reset)** and **F2 (Machine On)**.
7. Confirm the drive enables pick (S-ON to each TRA drive) and no drive faults show.

If Machine On fails: check the E-stop chain state (Ch. 12), drive fault inputs, and hydraulic pressure — all are preconditions.

Chapter 3 — Homing & Referencing

Axis position comes from the Tamagawa resolvers through the Mesa 7i49. Resolvers are absolute **within one revolution only**, so each axis must reference against its home switch after power-up.

1. Clear the table; ensure the Z axis has room to rise.
2. Home **Z first**, then X, then Y (keeps the head clear of fixtures).
3. Use the homing command in the GUI (Home All if configured; otherwise per-axis).
4. Watch that each axis moves *toward* its home switch (TB5.7–TB5.9). If any axis moves the wrong way, stop — the homing setup needs review (Vol 5 §6).
5. Verify machine zero against a known reference before trusting offsets.

Chapter 4 — Jogging & Manual Control

- Select jog increments (continuous / 1.0 / 0.1 / 0.01 / 0.001) before touching a jog key.
- Keyboard arrows / Page keys jog X, Y, Z in the standard LinuxCNC layout; the pendant (Ch. 10) may also be used.
- Overtravel limits (TB5.1–TB5.6) will halt motion, but they are the last resort — jog with your eyes on the machine, not the screen.
- Rapid-speed jogs only with the work area clear.

Chapter 5 — Spindle Operation

The Mitsubishi FR-SX drive receives its speed reference from the 7i97T analog output (TB3.15/16) and its run/direction/enable commands from the 7i84U (OUT0 FWD, OUT1 REV, OUT2 ENA).

- **M3 Sxxxx** — spindle forward at S RPM; **M4** — reverse; **M5** — stop.
- **Gear ranges:** the hydraulic Hi/Lo gearbox is commanded by `gear-hi-sol` / `gear-lo-sol` with confirmation switches PRS-10 / PRS-12. Allow gear shifts to complete (confirm input true) before loading the spindle.
- **At-speed:** feed moves that require the spindle wait on `spindle-at-speed` (FR-SX speed-reach output, IN13).
- **Orient:** spindle orient is required for tool change; do not command an ATC cycle unless orient completes.
- Do not command speed changes across a gear range mid-cut.

Chapter 6 — Tool Changes & the ATC / Magazine

The hydraulic ATC and BCD-coded magazine are retained from the original machine.

Automatic (M6 Tn): LinuxCNC sequences magazine position (BCD bits PRS-21...25), spindle orient, Z to tool-change position (zone confirms PRS-55/PRS-66), tool unclamp (SOL-10 via RLY-3/4, confirms PRS-8/PRS-9), arm swap (ATC motor FWD/REV), clamp, and retract.

Rules:

- Never reach into the changer while hydraulics are pressurized.
- If an ATC cycle stalls mid-sequence, do **not** jog axes — note the step, consult Vol 5 §8 (ATC recovery).

- Load tools in holders per the Tooling Manual (M00T002) with correct pull studs; a wrong stud can release under load.
- Keep the magazine tool table (`tool.tbl`) in sync with physical pockets — a mismatch is the most common ATC crash cause.

Chapter 7 — Running G-code Programs

1. Load the program (File → Open). The machine runs standard **EIA/ISO G-code** — Mazatrol conversational programs do **not** run directly; reprogram or post-process from CAM.
2. Set work offsets (G54...) by touching off each axis.
3. Set tool length offsets in the tool table for every tool used.
4. Verify with the preview plot; check extents against fixtures.
5. First run: **Single Block ON, rapid override down, feed override down**, hand on Feed Hold.
6. Production runs only after a proven first article.

Chapter 8 — Coolant, Air & Auxiliary Functions

Function	Command	Hardware
Flood coolant	M8 / M9	7i84U OUT11 → pump relay
Way lube	Auto/manual	7i84U OUT12; low-level switch IN25
Air blast	Custom M-code	7i97T SSR OUT0 → RLY-5 → SOL-62
Touch-sensor blast	Custom M-code	7i97T SSR OUT1 → RLY-6 → SOL-35
Tap coolant blast	Custom M-code	7i97T SSR OUT2 → RLY-7 → SOL-61
Alarm light/horn	Automatic on fault	7i84U OUT15

Coolant level (IN26) and lube level (IN25) warnings must be corrected promptly — the ways depend on the lube system (Vol 5 §3).

Chapter 9 — Feed Hold, Single Block & Cycle Control

- **Feed Hold** (panel button IN29 or GUI pause): decelerates axes, spindle keeps running. Resume with Cycle Start.
- **Single Block** (IN30): one block per Cycle Start press — use for first articles and recovery.
- **Cycle Start** (IN28 or GUI): starts/resumes the loaded program.
- **E-stop**: hardware chain — drives and hydraulics drop immediately. Use it whenever in doubt; recover via Ch. 2 power-up steps 6–7 and re-home.

Chapter 10 — Optional WHB04B Pendant

A WHB04B-style wireless USB pendant connects through LinuxCNC HAL (`pendant_whb04b.hal`), not through Mesa I/O. It provides axis select, step size, MPG wheel jogging, and cycle buttons. The pendant is enabled only after base-machine safety is proven. Pendant E-stop (if fitted) is a *soft stop* — the wall/panel E-stops remain the hardware chain.

Chapter 11 — Shutdown Sequence

1. Finish or abort the program; M5 spindle stop; M9 coolant off.
2. Move Z up and park X/Y at a convenient location.

3. Press E-stop (drops drive enables), exit LinuxCNC, shut down the PC.
4. Turn off the main disconnect if the machine will be idle.
5. Log any alarms or anomalies in the machine log for maintenance.

Chapter 12 — Operator Alarms & First Response

Symptom	Likely source	First response
Machine On refused	E-stop chain open (TB5.10)	Walk the chain: buttons, doors, overtravel finals
Joint following error	Resolver feedback / drive fault	Do not re-enable repeatedly; see Vol 5 §6
Drive fault input (X/Y/Z)	TRA drive alarm (IN10–IN12)	Read drive LED code; Vol 5 §4
Spindle won't start	Gear not confirmed, FR-SX fault (IN14), enable chain	Check gear confirm, VFD display; Vol 5 §7
ATC stalled	Interlock not satisfied	Note the step; Vol 5 §8 — never jog out blindly
Limit tripped	Overtravel	Jog off under limit-override per GUI procedure, then re-home

End of Volume 2. Continue to Volume 3 — Mesa Conversion & Installation.

Volume 3 — Mesa Conversion & Installation

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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- 6. Mesa 7i84U — Remote Field I/O
- 7. LinuxCNC Configuration Files
- 8. Bring-Up Procedure (8 Stages)
- 9. Verification Checklists
- 10. Rejected / Historical Architecture

Chapter 1 — Architecture Decision Summary

Selected: LinuxCNC PC → Ethernet (`hm2_eth` , static IP) → **Mesa 7i97T** analog servo controller, with a **Mesa 7i49** (plain) resolver interface for axis feedback and a **Mesa 7i84U** smart-serial field-I/O card near the existing green breakout PCB.

Why it fits this machine:

- Three analog servo axes + analog spindle command match the 7i97T exactly.
- The machine keeps its original **Tamagawa resolvers** — so feedback is resolver-to-digital (7i49), not quadrature encoder.
- Ethernet removes dependence on a PCIe slot and lets the PC sit in the old Mazatrol rack area.
- The 7i84U mounts next to the existing terminal strips, keeping ATC/hydraulic field wiring short.
- A **sister retrofit** (Mazak VQC 15/40, same Mitsubishi TRA drives and HD81-12S motors, plain 7i49 at 5 kHz) validates the approach — see `srdco/MazakVQC1540` on GitHub.

Chapter 2 — Hardware Stack & Bill of Boards

Item	Role	Notes
LinuxCNC control PC	Trajectory, GUI, HAL	Must pass latency test before install
Mesa 7i97T	X/Y/Z ±10 V analog commands, drive enables, spindle analog, limits/homes/E-stop, SSR outputs	Ethernet, static IP 192.168.1.121
Mesa 7i49 (plain)	Resolver-to-digital, X/Y/Z channels RES0–RES2	5 kHz excitation baseline; 7i49HV is contingency only
Mesa 7i84U	32 in / 16 out remote field I/O	Smart-serial from 7i97T TB4

Interposing relays RLY-1...7	Drive 100 VAC solenoid coils	With flyback / RC suppression
WHB04B pendant (optional)	MPG jogging	USB to PC, install last

Do not order a second smart-serial card unless the input count in `current_pin_authority.csv` is proven insufficient (project rule).

Chapter 3 — Control PC & Ethernet Setup

1. Install LinuxCNC (current stable) on the PC; run the **latency test** for at least an hour with typical desktop load. Servo-thread jitter must be acceptable before proceeding.
2. Configure the NIC on the Mesa subnet with a static host address (e.g. 192.168.1.1/24). No DHCP on this link.
3. Verify: `ping 192.168.1.121`, then `mesaflash --device 7i97t --addr 192.168.1.121 --readhmid`.
4. **Save the outputs** as `mesa_readhmid.txt` and, after firmware load, the HAL pin dump as `mesa_hal_pins.txt` — these replace every placeholder pin name in the HAL files.

Chapter 4 — Mesa 7i97T — Primary Motion Controller

Connector	Function on this machine
TB3	X/Z/Y analog velocity commands (AOUT0/1/2) + drive S-ON enables (ENA0/1/2); spindle speed AOUT3
TB4	Smart-serial link to 7i84U (absolute pins TB4.13–TB4.20)
TB5	Limits (IN0–IN5), homes (IN6–IN8), E-stop monitor (IN9), 24 V field power, SSR outputs 0–5
TB2	Axis encoder inputs NOT USED (resolver feedback); encoder.03 reserved for a machine spindle encoder if fitted

Confirm the correct firmware/bitfile exposes: 3+ analog channels, smart-serial on TB4, and the required GPIO. Note whether the board tag is `hm2_7i97t` or `hm2_7i97` in the real pin dump.

Axis-to-channel order is X=0, Z=1, Y=2 on TB3 (matches pin authority) — do not assume X/Y/Z order.

Chapter 5 — Mesa 7i49 — Resolver Feedback Interface

The most machine-specific part of this conversion. Read Vol 4 §3 alongside this chapter.

Facts & baseline:

- Resolvers: Tamagawa (RT-5XA-11 / RT-5XA-L1 documented; per-axis labels still to confirm). MS3108B 20–29P is a **connector shell**, not a resolver model.
- These are ~2:1 control-transformer resolvers; the 7i49 drives ~2 V RMS and expects ~1 V RMS sin/cos back — a natural match, hence **plain 7i49**.
- Excitation: **5 kHz** (spec 4.5 kHz; 7i49 offers 2.5/5/10). Firmware `num_resolvers` must expose 3 channels.

Mandatory verification sequence (drives disabled):

1. **Ohmmeter the winding pairs per axis before any power.** Original Meldas/TRA wiring may run the resolver backwards (excitation into stator). Expected once verified: rotor pair (likely RO1/RO2) → RESDRV+/-; two matched stator pairs → RESSIN and RESCOS.
2. Confirm the **7i49 is the sole excitation source** — nothing from the old drive/control may still touch the windings. (TRA drives close their velocity loop on the tachometer, so they don't need the resolver.)
3. Energize, **scope the sin/cos return** (~1 V RMS expected). Low → noise/sluggish response. Too hot → divider or W2 half-drive jumper (**field option, not default**). Far too weak at full drive → only then consider 7i49HV.
4. Rotate each axis by hand; confirm counts, direction, and derive the resolver scale (counts per machine unit) for the INI.

Chapter 6 — Mesa 7i84U — Remote Field I/O

- Mounts near the green breakout PCB; connects to 7i97T TB4 over smart-serial (full link map in Vol 4 §5). Shield drain lands at the **7i97T end only**.
- Field power banks: TB1 pins 1/2 (bank A → outputs 0–7, inputs 0–15), pins 3/4 (bank B → outputs 8–15, inputs 16–31). Confirm capacity and fusing from the HR-11F-24 24 V system before landing.
- Carries all ATC, magazine, hydraulic, coolant, lube, spindle-drive digital, panel, and alarm I/O — complete maps in Vol 4 §§8–9.
- Outputs that drive 100 VAC solenoid coils go through interposing relays RLY-1...RLY-7, never directly.

Chapter 7 — LinuxCNC Configuration Files

File	Purpose
mazak_vqc_20_40.ini	Machine, joints, spindle, HAL loading
mazak_vqc_20_40.hal	Main loader — loadrt hm2_eth board_ip=192.168.1.121
motion_7i97t.hal	Axis commands, resolver feedback nets, limits/homes/E-stop, enables
field_7i84u.hal	ATC, magazine, coolant, lube, hydraulic, panel I/O
postgui.hal, pendant_whb04b.hal	GUI extras, optional pendant
tool.tbl	Tool table — must match magazine pockets

These files are bring-up skeletons. Every `hm2_7i97t.*`/7i84U pin name, resolver scale, analog polarity, and NO/NC assumption is a placeholder until replaced with values from the real pin dump and cabinet measurements. Known stale nets are listed in the pin authority cleanup notes (e.g. tool clamp/unclamp confirm inputs moved from IN0/IN1 to IN15/IN16).

Chapter 8 — Bring-Up Procedure (8 Stages)

1. **Comms:** static IP, ping, mesaflash, `hm2_eth` loads, save readhmid + HAL pin dumps.
2. **24 V system:** trace HR-11F-24 (+S/+/-/-S, TOG, CNT, FG), P24/G24 distribution, remote sense, branch fusing, 0 V reference strategy.

3. **Resolvers (drives disabled):** full Chapter 5 sequence — ohmmeter, sole-excitation check, 5 kHz, scope, counts/direction/scale.
4. **Analog commands (drives inhibited):** verify 0 V at zero command; verify polarity per axis. A *reversed command* = runaway.
5. **One axis at a time:** low gain, low speed, small moves; tune; repeat per axis.
6. **Homes & limits:** prove switch logic before first homing cycle.
7. **Spindle:** verify FR-SX command mode (0–10 V vs bipolar — unresolved, see below), scaling at low RPM, run/direction/enable, zero-speed, at-speed, gear shifts, orient.
8. **ATC/hydraulics:** one output at a time, dry-run interlocks, no tool load.

Open decision to resolve during stage 7: spindle run/enable ownership — TB3 ENA3 candidate **conflicts** with the 7i84U FWD/REV/ENA plan (status HOLD_CONFLICT). Resolve before wiring either.

Chapter 9 — Verification Checklists

Pre-power: cabinet photos captured (checklist in repo `docs/`); safety chain traced and drawn; drive model/terminal labels recorded; FR-SX terminals recorded; resolver winding pairs ohm'd; 24 V fusing confirmed; nothing else on resolver windings.

Pre-motion: E-stop chain functional test (every button, door, overtravel final drops the contactors); estop-ext seen by LinuxCNC; drive enables verified per axis; zero-command verified; resolver counts/direction verified; following-error limits set conservatively.

Pre-ATC: every prox normal state measured (PRS-8/9/10/12/13/21–25, PHS-181/182, PRS-55/66); every solenoid coil voltage/current measured; relay/suppression fitted; hydraulic pressure switch proven; magazine BCD decode verified against physical pockets; dry-run with no tools.

Chapter 10 — Rejected / Historical Architecture

An earlier plan used a tower PC with a **6i25 PCIe** card, **7i77** analog servo card, **7i84** field I/O, and optional 7i85/7i85S. It was **superseded** by the 7i97T + 7i84U Ethernet architecture. Any document or workbook tab referencing 7i77 terminals or Magnescale quadrature encoders is **legacy** — do not wire from it. (Legacy tabs are clearly separated in the wiring master workbook.)

End of Volume 3. Continue to Volume 4 — Wiring Diagrams & I/O Reference.

Volume 4 — Wiring Diagrams & I/O Reference

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

Document MZK-LCNC-004 · Rev A · July 2026

⚠ **STATUS:** Diagrams and tables reflect the **current pin authority** (planning stage). Rows marked **PLAN_VERIFY** / **ACCEPTED_VERIFY** must be traced in the cabinet before landing wires. The original Mazak schematic set **41434WB** remains authoritative for machine-side wiring.

Table of Contents

- 1. System Architecture Diagram
- 2. 7i97T TB3 — Servo Analog Commands & Enables
- 3. 7i49 — Resolver Wiring
- 4. 7i97T TB5 — Limits, Homes, E-stop & SSR Outputs
- 5. 7i97T TB4 ↔ 7i84U — Smart-Serial Link
- 6. Spindle Control — Mitsubishi FR-SX
- 7. E-stop / Safety Chain Concept
- 8. 7i84U TB1 — Complete Input Map
- 9. 7i84U TB2 — Complete Output Map
- 10. Legend, Status Codes & Field Devices

§1 — System Architecture

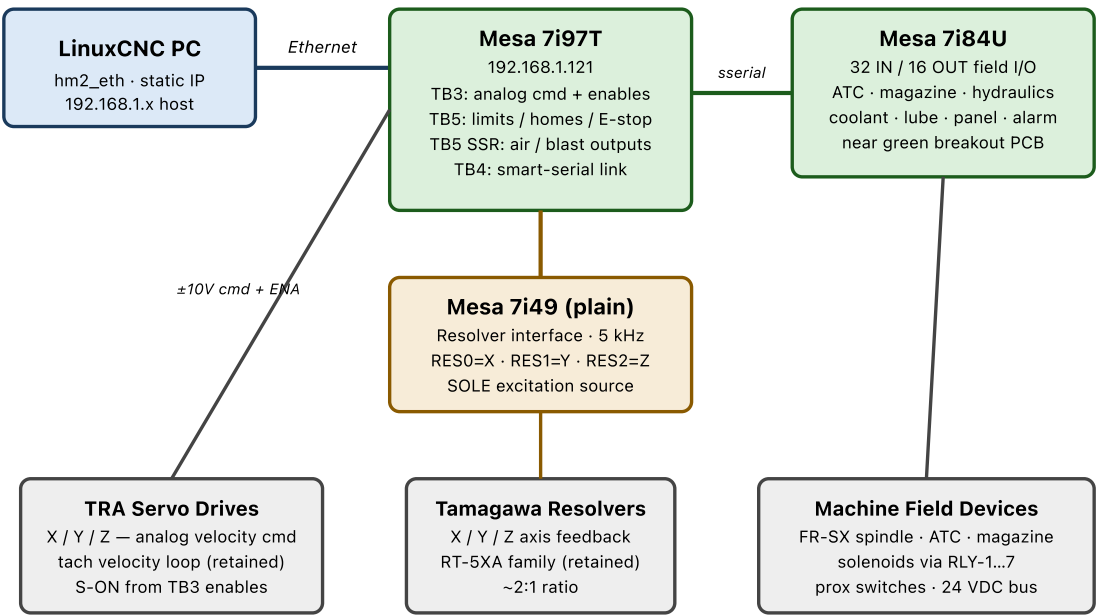


Figure 4-1. Control topology. The PC talks only to the 7i97T; the 7i84U hangs off smart-serial; the 7i49 provides resolver channels. All machine power electronics are retained.

§2 — 7i97T TB3: Servo Analog Commands & Enables

Note the channel order: **X = channel 0**, **Z = channel 1**, **Y = channel 2**.

TB3 pins	Signal	HAL net	Goes to	Status
----------	--------	---------	---------	--------

TB3.1 / TB3.2	ENA0– / ENA0+	x-enable	X servo S-ON	PLAN_VERIFY
TB3.3 / TB3.4	GND / AOUT0	x-vel-cmd	X drive analog cmd	PLAN_VERIFY
TB3.5 / TB3.6	ENA1– / ENA1+	z-enable	Z servo S-ON	PLAN_VERIFY
TB3.7 / TB3.8	GND / AOUT1	z-vel-cmd	Z drive analog cmd	PLAN_VERIFY
TB3.9 / TB3.10	ENA2– / ENA2+	y-enable	Y servo S-ON	PLAN_VERIFY
TB3.11 / TB3.12	GND / AOUT2	y-vel-cmd	Y drive analog cmd	PLAN_VERIFY
TB3.13 / TB3.14	ENA3– / ENA3+	spindle-enable	VFD FWD/RUN <i>candidate</i>	HOLD_CONFLICT
TB3.15 / TB3.16	GND / AOUT3	spindle-speed-cmd	FR-SX V-IN speed ref	PLAN_VERIFY

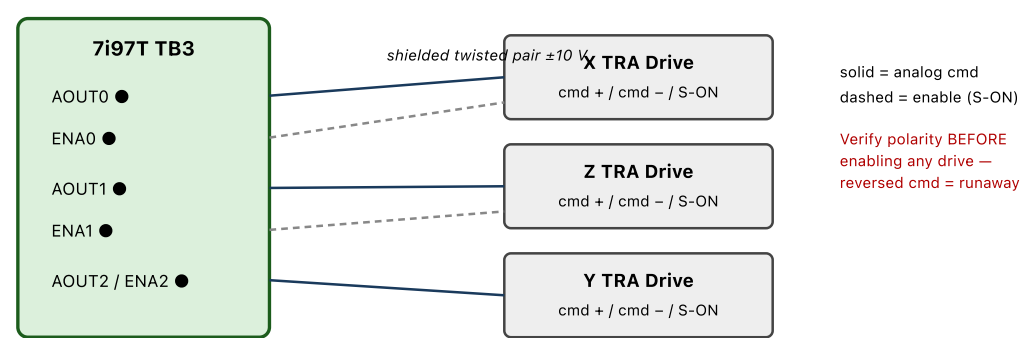
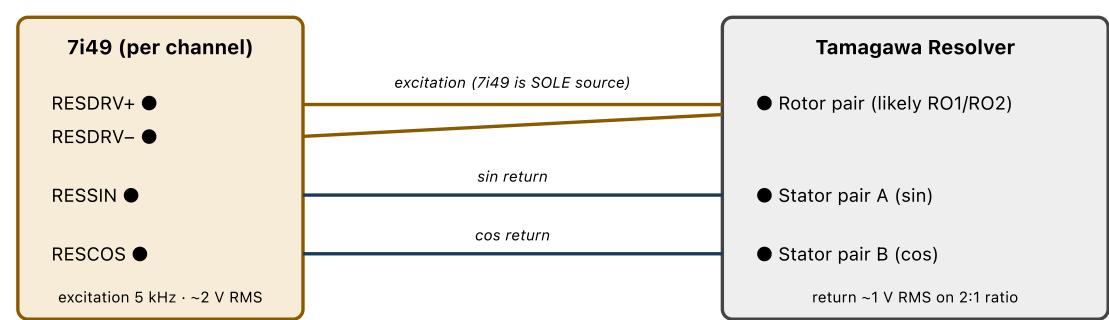


Figure 4-2. TB3 to TRA drives. Use shielded twisted pairs for analog; keep them away from contactor/motor power runs.

§3 — 7i49 Resolver Wiring

Channel	Axis	HAL net	Field device	Status
RES0	X	x-pos-fb	Tamagawa X resolver	PLAN_VERIFY
RES1	Y	y-pos-fb	Tamagawa Y resolver	PLAN_VERIFY
RES2	Z	z-pos-fb	Tamagawa Z resolver	PLAN_VERIFY



OHMMETER THE WINDING PAIRS BEFORE POWER — original Meldas/TRA wire names may be backwards

Figure 4-3. Resolver channel wiring. All pairs shielded; shield termination strategy to be finalized (Vol 3 §5). W2 half-drive jumper is a field-verification option only. Escalate to 7i49HV only if the return is far too weak at full drive.

§4 — 7i97T TB5: Limits, Homes, E-stop & SSR Outputs

TB5 pin	GPIO	HAL net	Field device	Status
TB5.1 IN0	gpio.008	limit-x-plus	X + overtravel	ACCEPTED_VERIFY
TB5.2 IN1	gpio.009	limit-x-minus	X – overtravel	ACCEPTED_VERIFY
TB5.3 IN2	gpio.010	limit-y-plus	Y + overtravel	ACCEPTED_VERIFY
TB5.4 IN3	gpio.011	limit-y-minus	Y – overtravel	ACCEPTED_VERIFY
TB5.5 IN4	gpio.012	limit-z-plus	Z + overtravel	ACCEPTED_VERIFY
TB5.6 IN5	gpio.013	limit-z-minus	Z – overtravel	ACCEPTED_VERIFY
TB5.7 IN6	gpio.014	home-x	X home switch	ACCEPTED_VERIFY
TB5.8 IN7	gpio.015	home-y	Y home switch	ACCEPTED_VERIFY
TB5.9 IN8	gpio.016	home-z	Z home switch	ACCEPTED_VERIFY
TB5.10 IN9	gpio.017	estop-ext	E-stop chain monitor	ACCEPTED_VERIFY
TB5.11	—	—	24 V field common (GND)	PLAN_VERIFY
TB5.12	—	—	+VFIELD 24 V supply	PLAN_VERIFY

TB5 SSR outputs (overflow relay drivers — coils are 100 VAC, always via interposing relay):

TB5 pins	GPIO	HAL net	Load	Status
TB5.13–14 SSR OUT0	gpio.020	air-blast	RLY-5 → SOL-62	PLAN_VERIFY
TB5.15–16 SSR OUT1	gpio.021	touch-sensor-blast	RLY-6 → SOL-35	PLAN_VERIFY
TB5.17–18 SSR OUT2	gpio.022	tap-coolant-blast	RLY-7 → SOL-61	PLAN_VERIFY
TB5.19–24 SSR OUT3–5	gpio.023–025	—	Spare — keep unassigned	SPARE

§5 — 7i97T TB4 ↔ 7i84U Smart-Serial Link

All rows **ACCEPTED**. Use absolute TB4 pin numbers. TX/RX labels are from each card's own perspective (7i97T TX → 7i84U RX).

7i97T TB4	Signal	7i84U RJ45 pin	Note
TB4.13	GND	4	ground A
TB4.14	GND	5	ground B — shield drain at 7i97T end only
TB4.15	RX+	6 (TX+)	verify continuity before power
TB4.16	RX–	3 (TX–)	verify continuity before power
TB4.17	TX+	2 (RX+)	verify continuity before power
TB4.18	TX–	1 (RX–)	verify continuity before power
TB4.19	+5V	7	serial power — confirm w/ 7i84U manual
TB4.20	+5V	8	serial power — confirm w/ 7i84U manual

§6 — Spindle Control: Mitsubishi FR-SX

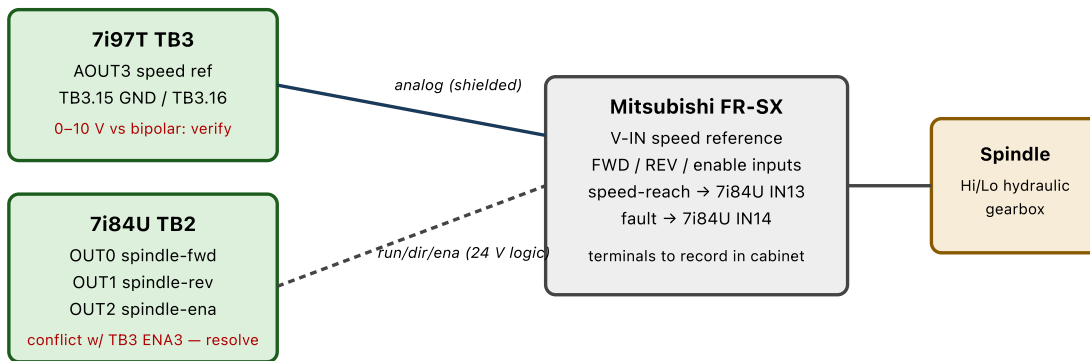


Figure 4-4. Spindle control. Gear shift: `gear-hi-sol` (OUT7 → RLY-1 → SOL-13) and `gear-lo-sol` (OUT8 → RLY-2 → SOL-12), confirmed by PRS-10 (IN17) / PRS-12 (IN18).

§7 — E-stop / Safety Chain Concept

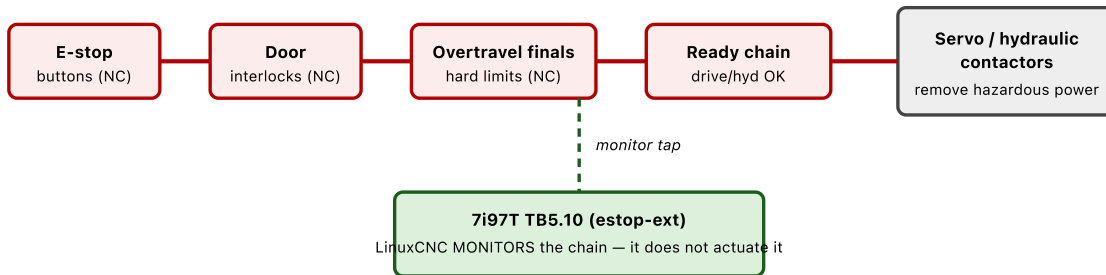


Figure 4-5. Series hardware chain drops the contactors directly. The Mesa input is a monitor tap only. Exact original chain wiring must be traced from schematic 41434WB and the cabinet before rewiring (Vol 3 §9).

§8 — 7i84U TB1: Complete Input Map (32 inputs)

Field power: TB1 pins 1/2 = bank A (+/–, feeds OUT0–7 & IN0–15); pins 3/4 = bank B (feeds OUT8–15 & IN16–31). Status is PLAN_VERIFY unless noted.

IN	HAL net	Field device / meaning
IN0	atc-y-zone	PRS-55 — Y axis in ATC zone
IN1	atc-z-zone	PRS-66 — Z axis in ATC zone
IN2	mag-tool-avail	PHS-181 — magazine tool available
IN3	spindle-tool-avail	PHS-182 — spindle tool available
IN4–IN9	—	Spare (keep for future field signals)
IN10	x-drive-fault	X TRA drive fault relay
IN11	y-drive-fault	Y TRA drive fault relay
IN12	z-drive-fault	Z TRA drive fault relay
IN13	spindle-at-speed	FR-SX speed-reach output
IN14	spindle-fault	FR-SX fault output
IN15	tool-clamped	PRS-9 — tool clamp confirm

IN16	tool-unclamped	PRS-8 — tool unclamp confirm
IN17	gear-hi-conf	PRS-10 — gear high confirm
IN18	gear-lo-conf	PRS-12 — gear low confirm
IN19	mag-bcd-bit0	PRS-21 — magazine BCD bit 0
IN20	mag-bcd-bit1	PRS-22 — magazine BCD bit 1
IN21	mag-bcd-bit2	PRS-23 — magazine BCD bit 2
IN22	mag-bcd-bit3	PRS-24 — magazine BCD bit 3
IN23	mag-bcd-bit4	PRS-25 — magazine BCD bit 4
IN24	door-interlock	Door interlock switch (state TBD after measurement)
IN25	lube-level	Lube level switch
IN26	coolant-level	Coolant level switch
IN27	hydraulic-ok	Sanwa SPS-8T-PC-20 pressure switch
IN28	cycle-start-pb	Cycle Start pushbutton (<i>optional panel reuse</i>)
IN29	feed-hold-pb	Feed Hold pushbutton (<i>optional panel reuse</i>)
IN30	single-block	Single Block selector (<i>optional panel reuse</i>)
IN31	servo-ready	Servo drives ready relay contact

Magazine BCD bits decode via `bcd2s` or a custom mux in HAL.

§9 — 7i84U TB2: Complete Output Map (16 outputs)

OUT	HAL net	Load	Notes
OUT0	spindle-fwd	FR-SX forward input	resolve vs TB3 ENA3 candidate
OUT1	spindle-rev	FR-SX reverse input	resolve vs TB3 ENA3 candidate
OUT2	spindle-ena	FR-SX enable input	resolve vs TB3 ENA3 candidate
OUT3– OUT5	—	Spare	drive S-ON handled at 7i97T TB3
OUT6	z-brake-rel	N1J-L2-201 Z brake release	flyback/suppression required
OUT7	gear-hi-sol	RLY-1 → SOL-13 (Fujikoshi hydraulic valve)	measure coil V before relay pick
OUT8	gear-lo-sol	RLY-2 → SOL-12 (Fujikoshi hydraulic valve)	measure coil V before relay pick
OUT9	tool-clamp-sol	RLY-3 → SOL-10	confirm single-valve vs dual-coil
OUT10	tool-unclamp-sol	RLY-4 → SOL-10	confirm single-valve vs dual-coil
OUT11	flood-coolant	Coolant pump relay	may need interposing relay
OUT12	lube-on	Lube pump relay	may need interposing relay
OUT13	atc-fwd	ATC motor forward relay	never energize before interlocks proven

OUT14	atc-rev	ATC motor reverse relay	never energize before interlocks proven
OUT15	alarm-out	Alarm light / horn (<i>optional</i>)	confirm load

§10 — Legend, Status Codes & Field Devices

Authority status codes (from `current_pin_authority.csv`):

Code	Meaning
ACCEPTED	Settled; land per table (still continuity-check first)
ACCEPTED_VERIFY	Assignment accepted; field device/normal state still to verify
PLAN_VERIFY	Engineered plan; trace in cabinet before landing wires
HOLD_CONFLICT	Two plans conflict — resolve before wiring either
RESERVED_VERIFY	Held for a possible device (e.g. spindle encoder)
OPTIONAL_VERIFY	Nice-to-have panel reuse
SPARE	Keep unassigned

Field-device prefix legend: PRS = proximity/position switch · PHS = photo/hall sensor · SOL = hydraulic/pneumatic solenoid · RLY = interposing relay (retrofit-added) · TRA = Mitsubishi servo drive family · FR-SX = Mitsubishi spindle drive.

Drawing cross-reference: machine-side power, motor, and original interlock wiring — Mazak schematic set **41434WB** (6/1984). Original PLC signal meanings — **YM2V39L** ladder manual and its extracted element list (385 signals).

End of Volume 4. Continue to Volume 5 — Maintenance & Troubleshooting.

Volume 5 — Maintenance & Troubleshooting

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

Document MZK-LCNC-005 · Rev A · July 2026

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1. Maintenance Philosophy & Original Manual Cross-Reference
2. Daily / Weekly / Monthly Schedules
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5. Mesa Board & LinuxCNC Health
6. Troubleshooting — Motion
7. Troubleshooting — Spindle
8. Troubleshooting — ATC / Magazine
9. Troubleshooting — Mesa Comms / HAL
10. Spare Parts & Sourcing

Chapter 1 — Maintenance Philosophy & Cross-Reference

The machine's mechanics are unchanged by the retrofit, so the **original Mazak Maintenance Manual (413m033)** remains the authority for ways, ballscrews, spindle bearings, hydraulics, and lubrication intervals. This volume covers what the original manual cannot: the Mesa/LinuxCNC control layer, and the boundary between old iron and new electronics.

Topic	Authority
Mechanical maintenance, lube points, hydraulic specs	413m033 (Maintenance Manual)
Machine electrical (motors, transformer, contactors)	41434WB (Electrical Circuit Diagram)
Original interlock logic / signal meanings	YM2V39L (PLC Ladders) + element list CSV
Spare mechanical parts	413LE02A000 (Parts List)
Control layer (PC, Mesa, HAL, wiring to Mesa)	This manual set (Vols 3–4)

Chapter 2 — Daily / Weekly / Monthly Schedules

Daily (operator):

- Check way-lube reservoir; verify `lube-level` shows OK.
- Check hydraulic pressure gauge and `hydraulic-ok` state.
- Check coolant level and condition.
- Listen during first warm-up: spindle, hydraulic pump, ATC noises.
- Clear chips from way covers, ATC arm area, and door tracks.

Weekly:

- Grease/oil points per 413m033 schedule.
- Inspect resolver and analog cable runs for chafing or oil intrusion.

- Verify E-stop chain: press each button; confirm contactors drop and LinuxCNC shows estop.
- Check cabinet filters/fans; clean PC intake.

Monthly:

- Backup the LinuxCNC config directory and `tool.tbl` (copy to the GitHub repo + OneDrive).
- Inspect terminal strips for loosened screws (thermal cycling).
- Check drive fault history (TRA LED history where available; LinuxCNC logs).
- Test limit switches by slow jog toward each; confirm halt.
- Verify following-error trends per axis haven't degraded (halscope snapshot).

Chapter 3 — Lubrication & Hydraulics

- Follow 413m033 for oil types, quantities, and intervals — the retrofit changed nothing mechanical.
- The lube pump is now controlled via 7i84U OUT12 (`lube-on`); low level appears on IN25. If lube logic misbehaves, verify the HAL timer settings against the original ladder behavior (YM2V39L) before suspecting the pump.
- Hydraulics power the ATC, tool clamp, gear shift, and (via N1J-L2-201) the Z brake circuit. Pressure OK is monitored by the Sanwa SPS-8T-PC-20 on IN27. **No ATC or gear operation should ever be commanded without pressure OK.**

Chapter 4 — Servo Drives & Resolver Feedback Health

- **TRA drives:** keep vent paths clear; record any fault LED codes with a photo before resetting. Drive faults land on 7i84U IN10–IN12.
- **Tachometers:** the drives close their velocity loop on tachs — brush/commutator wear shows as speed instability that no LinuxCNC tuning will fix.
- **Resolvers:** sealed and robust, but connectors (MS-type circular shells) and cabling are the weak point. Symptoms of feedback trouble: position noise at standstill, following errors that scale with speed, or an axis that "hunts."
- Annually (or after any cabinet work): re-scope resolver sin/cos return amplitude and compare to the bring-up baseline recorded in Vol 3 §5.

Chapter 5 — Mesa Board & LinuxCNC Health

- Keep the 7i97T/7i49/7i84U boards free of chip dust and coolant mist; verify cabinet sealing.
- After any LinuxCNC update: re-run the latency test, and diff the HAL pin dump against `mesa_hal_pins.txt` before trusting the config.
- Keep firmware/bitfile versions recorded in `mesa/mesa_firmware_checklist.md` ; never flash new firmware mid-project without re-verifying pin names.
- Keep a bootable clone or image of the control PC drive — the PC is now a machine spare part.

Chapter 6 — Troubleshooting — Motion

Symptom	Checks (in order)
Joint following error on enable	Drive enabled? S-ON wiring TB3; drive fault input; resolver counts moving with hand rotation?
Following error at speed	Resolver return amplitude; tach health; drive gain; ferror settings vs actual tuning

Axis runs away on enable	Stop. Analog polarity reversed — re-verify Vol 3 §8 step 4 before re-enabling
Position noise at standstill	Resolver shield/ground; cable routing near power; return amplitude too low
Wrong direction on jog	Reverse resolver scale sign AND verify limits still protect correct ends
Homing goes wrong way	Home switch wiring (TB5.7–9), HOME_SEARCH_VEL sign in INI
Axis dead, no fault	Enable chain, TB3 continuity, drive power, contactor state

Chapter 7 — Troubleshooting — Spindle

Symptom	Checks
Won't start	E-stop chain; gear confirm (IN17/IN18 — must have exactly one true); FR-SX fault (IN14); enable path (resolve TB3-vs-7i84U ownership per Vol 3 §8)
Wrong speed	AOUT3 scaling; FR-SX command mode (0–10 V vs bipolar); gear range vs commanded S
No at-speed signal	IN13 wiring; FR-SX speed-reach parameter
Trips under load	Gear range too high for cut; FR-SX fault code on drive display
Won't orient	Orient logic depends on spindle feedback — verify encoder.03 reservation status and orient hardware per 41434WB

Chapter 8 — Troubleshooting — ATC / Magazine

Golden rule: never jog axes to "free" a stalled ATC. Identify the stalled step first.

Symptom	Checks
M6 refused	Tool-avail inputs (IN2/IN3); zone confirms PRS-55/66 (IN0/IN1); hydraulic OK (IN27); orient complete
Unclamp fails	SOL-10 via RLY-3/4 energizing? PRS-8/9 confirm switches; hydraulic pressure; dual-coil vs single-valve behavior (Vol 4 §9 note)
Magazine wrong pocket	BCD inputs IN19–23 vs physical pocket — test decode with bcd2s watch; check PRS-21...25 alignment
Arm stalls mid-swap	Which interlock dropped? Record all IN0–IN3, IN15/IN16 states before touching anything
Gear/ATC solenoid chatter	Coil voltage sag; relay contact rating; suppression missing

Recovery: restore hydraulic pressure, resolve the blocking input, and complete the cycle in the direction it was moving where possible. The original ladder (YM2V39L) documents the intended sequence — consult it

before manual intervention.

Chapter 9 — Troubleshooting — Mesa Comms / HAL

Symptom	Checks
LinuxCNC won't start / hm2_eth error	ping 192.168.1.121; NIC static IP; cable; mesaflash --readhmid
Random real-time delays	Latency test; disable CPU power management; NIC IRQ sharing
7i84U pins missing	Smart-serial link (Vol 4 §5) continuity; TB4 wiring; sserial error counters in HAL
Input never changes	Field power bank A/B present? Correct bank for that input number? Sensor 24 V supply
Output won't fire	Watch the HAL pin first (halcmd show pin); then bank power; then relay coil
Everything dead after config edit	Diff against the repo copy — restore from Git

Chapter 10 — Spare Parts & Sourcing

- **Mechanical:** source from the Parts List 413LE02A000 breakdown; Mazak can quote by publication + serial number.
- **Drives/motors:** Mitsubishi TRA / HD81-12S parts are obsolete — used-market items. The sister build (srdco/MazakVQC1540) community is a practical sourcing lead.
- **Mesa boards:** keep a spare of each board type once in production; store the exact firmware/bitfile with it.
- **Recommended shelf spares:** interposing relays (RLY-1...7 type), 24 V supply (HR-11F-24 equivalent), prox switch equivalents for PRS types, fuses per branch list, control PC image on USB.

End of Volume 5. Continue to Volume 6 — Reference, Glossary & Index.

Volume 6 — Reference, Glossary & Index

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

Document MZK-LCNC-006 · Rev A · July 2026

Table of Contents

- 1. Original Mazak Manual Library
- 2. Key Project Files
- 3. External References
- 4. Glossary of Terms
- 5. Alphabetical Index (full set)

Chapter 1 — Original Mazak Manual Library

All originals live in `OneDrive-Personal/OneDrive Docs/Mazak/Manuals_SN060231/` (synced locally and shared via OneDrive). Source: Mazak Corporation secure download, 2025-10-13.

Publication	Title	Folder	Retrofit relevance
413S038	Operating Manual — VQC-20/40 & 20/50	01_Operating	Machine handling basics; Mazatrol operation now historical
m00s075	Operating Manual — CAM M-2	01_Operating	Historical (Mazatrol UI retired)
413m033	Maintenance Manual — VQC-20/40 & 20/50	02_Maintenance	Active authority — mechanics, lube, hydraulics
41434WB	Electrical Circuit Diagram (M-2), 6/1984	02_Maintenance	Active authority — machine-side wiring; primary retrofit source
YM2V39L	PLC Ladder Diagrams	02_Maintenance	Active — interlock behavior to reproduce in HAL (385-signal element list extracted)
PAREXM210E	Parameter List & Explanation — M-2	03_Parameters	Historical + useful for decoding legacy setup values
M00P111	Programming Manual — CAM M-2 (EIA/ISO)	04_Supplements	G-code reference for legacy programs
m00p135	Programming Manual — M-2E	04_Supplements	Reference only — this machine is M-2, not M-2E
M00P018	Programming Manual — Mazatrol CAM 3D	04_Supplements	Option — verify if installed on SN 060231
M00T002	Tooling Manual — Mazak Machining Center	04_Supplements	Active — holders and pull studs

413LE02A000	Parts List — VQC 20/40 & 50	05_Parts	Active — spare parts sourcing
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Also in that folder: `servo_amp_analysis.pdf/.md` , `servo_topology_p082.png` , `terminal_unit_detail_p085.png` , cabinet photos (`photos_2026-07/`).

Chapter 2 — Key Project Files

File	Path	Role
Architecture decision	GitHub/Github LinuxCNC/docs/architecture_decision.md	Why 7i97T + 7i49 + 7i84U
Project status / TODO	GitHub/Github LinuxCNC/docs/project_status.md	Live checklist and bring-up order
Pin authority	GitHub/Github LinuxCNC/mesa/current_pin_authority.csv	The wiring truth table (supersedes signal_map.csv)
Wiring master workbook	GitHub/Github LinuxCNC/wiring/Mazak_Wiring_Master_7i97T_7i49_7i84U.xlsx	Consolidated workbook; legacy tabs clearly separated
I/O workbook	GitHub/Github LinuxCNC/bom/Mazak_VQC_20-40_Retrofit_IO_Workbook.xlsx	Original I/O planning source
HAL/INI skeletons	GitHub/Github LinuxCNC/linuxcnc/	Bring-up configs (placeholders until pin dump)
Firmware checklist	GitHub/Github LinuxCNC/mesa/mesa_firmware_checklist.md	Facts to collect before locking HAL names
PLC element list	Manuals_SN060231/02_Maintenance/YM2V39L_element_list.csv	385 legacy signals: 179 X-inputs, 206 Y-outputs

Chapter 3 — External References

- Mesa 7i49 manual (excitation, W2 jumper, RESDRV/RESSIN/RESCOS): mesanet.com/pdf/motion/7i49man.pdf
- Sister retrofit — `srcdco/MazakVQC1540` (VQC 15/40, same TRA drives/HD81-12S motors, plain 7i49 @ 5 kHz), incl. the complete 2017-05-01 reference package
- LinuxCNC forum: servo PID tuning thread (VQC 15/40) and the owner's build thread "Mesa conversion for a Mazak VQC 20/40 M2 mill"
- Mitsubishi TRA-31 drive manual; Meldas YM2 / Mazatrol M2 maintenance manual (Mitsubishi FA knowledge base)

Chapter 4 — Glossary

Term	Meaning
ATC	Automatic Tool Changer — hydraulic arm + magazine on this machine
BCD	Binary-Coded Decimal — how magazine pocket position is reported (PRS-21...25)
Bring-up	Staged first-power procedure proving each subsystem safely (Vol 3 §8)
ENA / S-ON	Drive enable signal from 7i97T TB3 to a TRA servo drive
E-stop chain	Series-wired hardware circuit that drops contactors directly
FR-SX	Mitsubishi spindle drive taking an analog speed reference
HAL	Hardware Abstraction Layer — LinuxCNC's signal-wiring layer
hm2_eth	LinuxCNC driver for Ethernet-connected Mesa FPGA boards
INI	LinuxCNC machine configuration file
Interposing relay	Small relay (RLY-1...7) between a Mesa output and a heavy/100 VAC coil
Mazatrol M-2	Original conversational CNC control (retired by this retrofit)
mesaflash	Utility to query/flash Mesa boards (<code>--readhmid</code>)
PLAN_VERIFY	Pin-authority status: engineered plan, must be traced before wiring
PRS / PHS	Proximity / photo-sensor switch designators from the original ladder
Resolver	Analog rotary position transducer; absolute within one rev; read by the 7i49
RESDRV / RESSIN / RESCOS	7i49 excitation drive and sin/cos return terminals
RLY	Retrofit-added interposing relay
Smart-serial (sserial)	Mesa's serial field-bus linking 7i97T TB4 to the 7i84U
SOL	Hydraulic/pneumatic solenoid valve designator
SSR	Solid-state relay output (7i97T TB5.13–24)
Tachometer	Analog speed sensor closing the TRA drive's velocity loop
TRA	Mitsubishi analog servo drive family retained in this machine
W2 jumper	7i49 half-excitation option — field-verification use only
WHB04B	Wireless USB MPG pendant (optional)

Chapter 5 — Alphabetical Index

References: **Volume** · chapter/§. Example: V4 §3 = Volume 4, section 3.

A

- Air blast (SOL-62): V2 c8; V4 §4
- Alarm output: V2 c8; V4 §9
- Alarms, operator response: V2 c12
- Analog command wiring: V3 c8; V4 §2
- Analog polarity / runaway hazard: V1 c6; V3 c8; V5 c6
- Architecture (selected): V3 c1; V4 §1
- Architecture (rejected 6i25/7i77): V3 c10
- ATC operation: V2 c6 · troubleshooting: V5 c8 · wiring: V4 §§8–9

B

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